1. What is the thinnest film of MgF_2 (n=1.39) on glass that produces a strong reflection for orange light with a wavelength of 600 nm?

2. Suppose you have a lens primarily used for 700-nm red light. What is the second thinnest coating of magnesium fluoride (n = 1.41) that will be non-reflective for this wavelength?

3. A sheet of glass is coated witha 500-nm-thick layer of oil (n = 1.42)

- **A.** For what *visible* wavelengths of light do the reflected waves interfere constructively?
- **B.** For what *visible* wavelengths of light do the reflected waves interfere destructively?
- **C.** What is the color of reflected light? What is the color of transmitted light?

4. Scientists are testing a transparent material whose index of refraction for visible light varies with wavelength as $n = \frac{30}{\sqrt{\lambda}}$, where λ is in nm. If a 295-nm-thick coating is placed on glass (n = 1.5) for what visible wavelengths will the reflected light have maximum constructive interference?

5. A michelson interferometer has two equal arms. A mercury light of wavelength 546 nm is used for the interferometer and stable fringes are found. One of the arms is moved by 1.5 μm. How many fringes will move past the observing field?

6. When the moveable mirror of a Michelson Interferometer is moved by 2.4×10^{-5} m, 90 fringes pass by a point on the observing screen. What is the wavelength of light used?